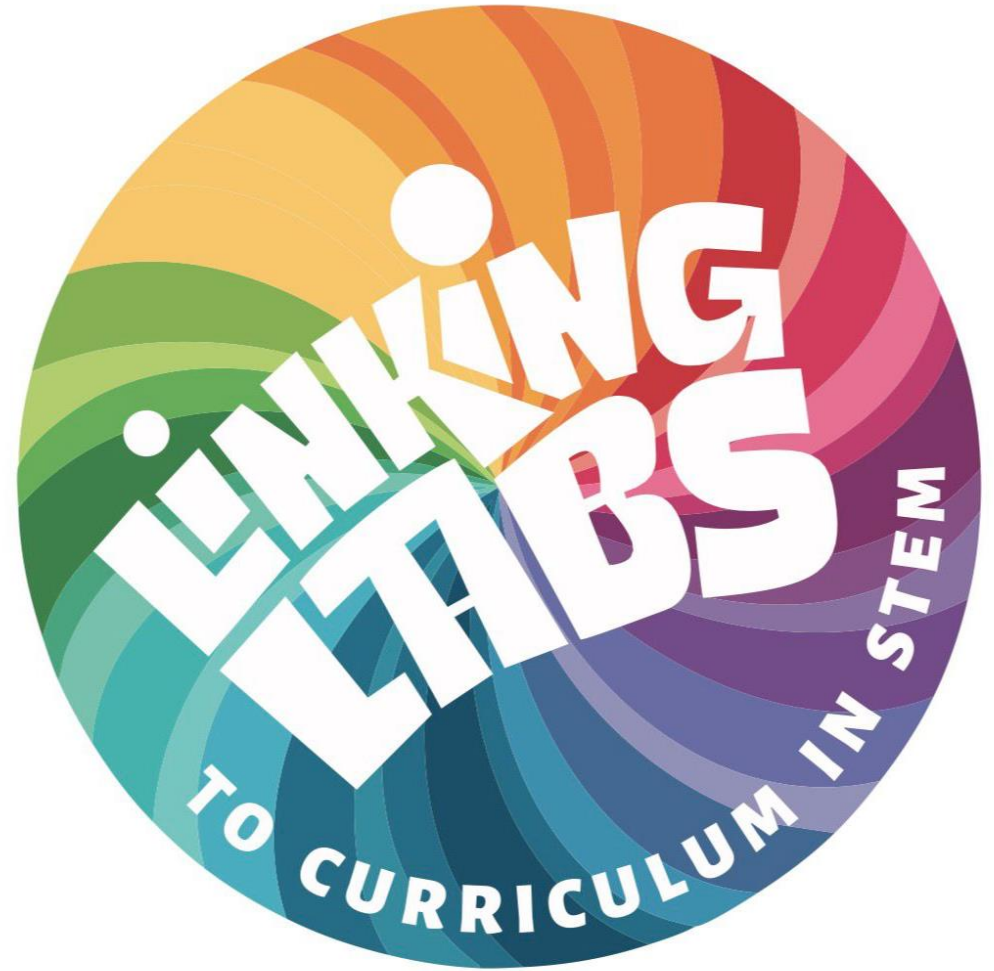


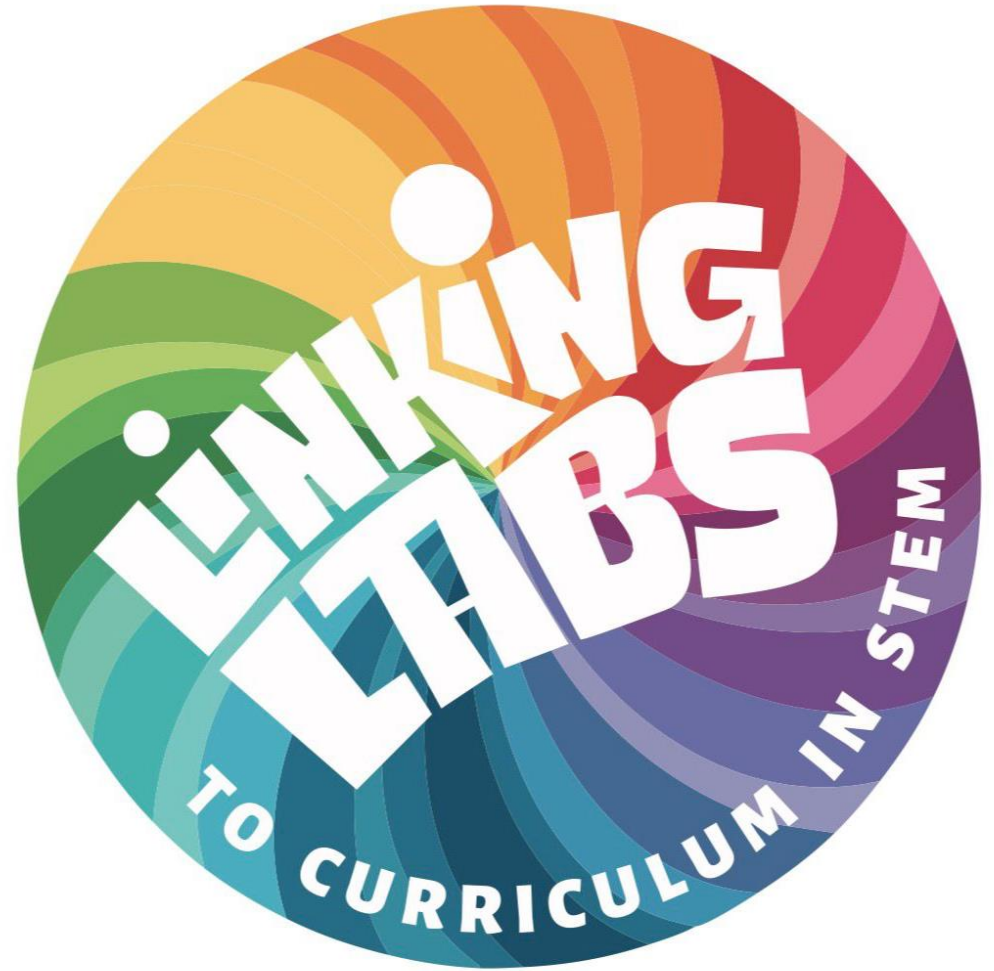
Today's agenda

- First activity: What are these folks doing?
- Second activity: "ouch I cut my finger", what is going to happen to me??
- Third activity: Building a rainbow



Today's agenda

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What are these folks doing?

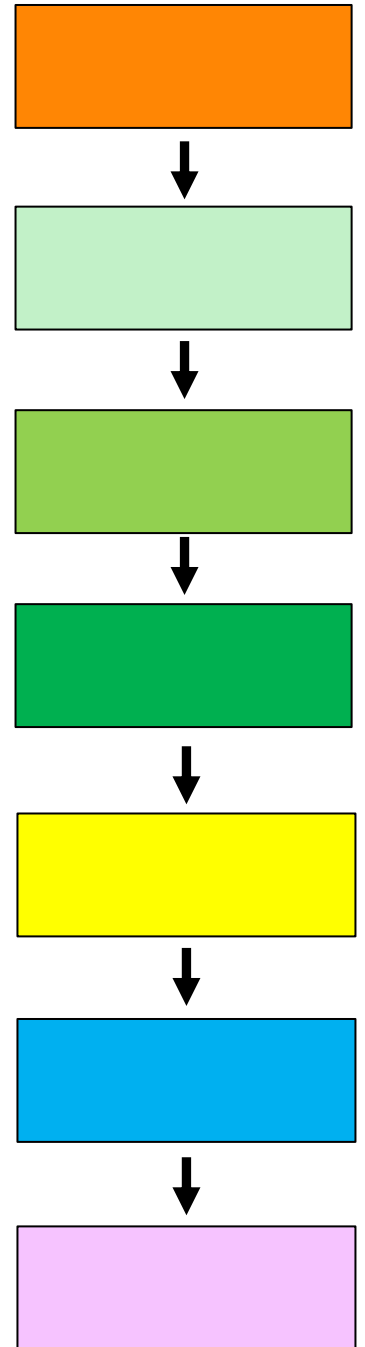


They are applying
the scientific method



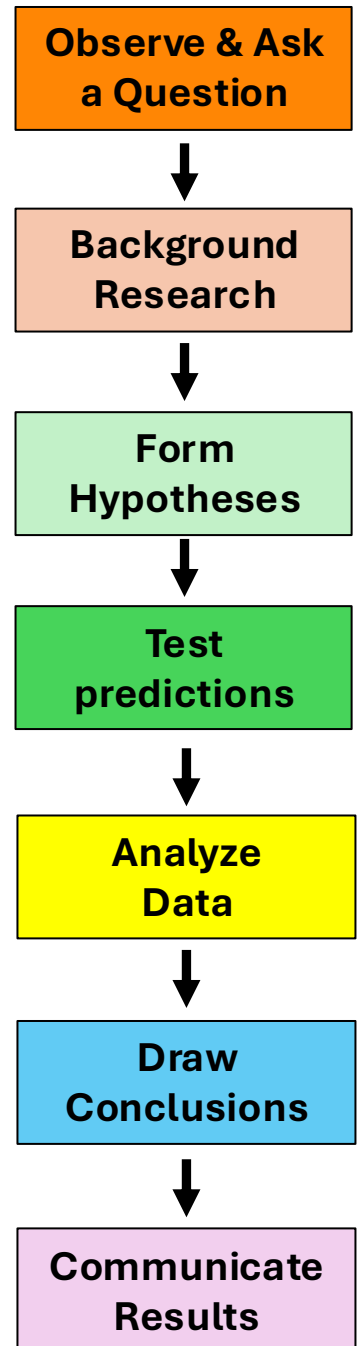
The Scientific Method

Do we remember the steps?
We start by.....



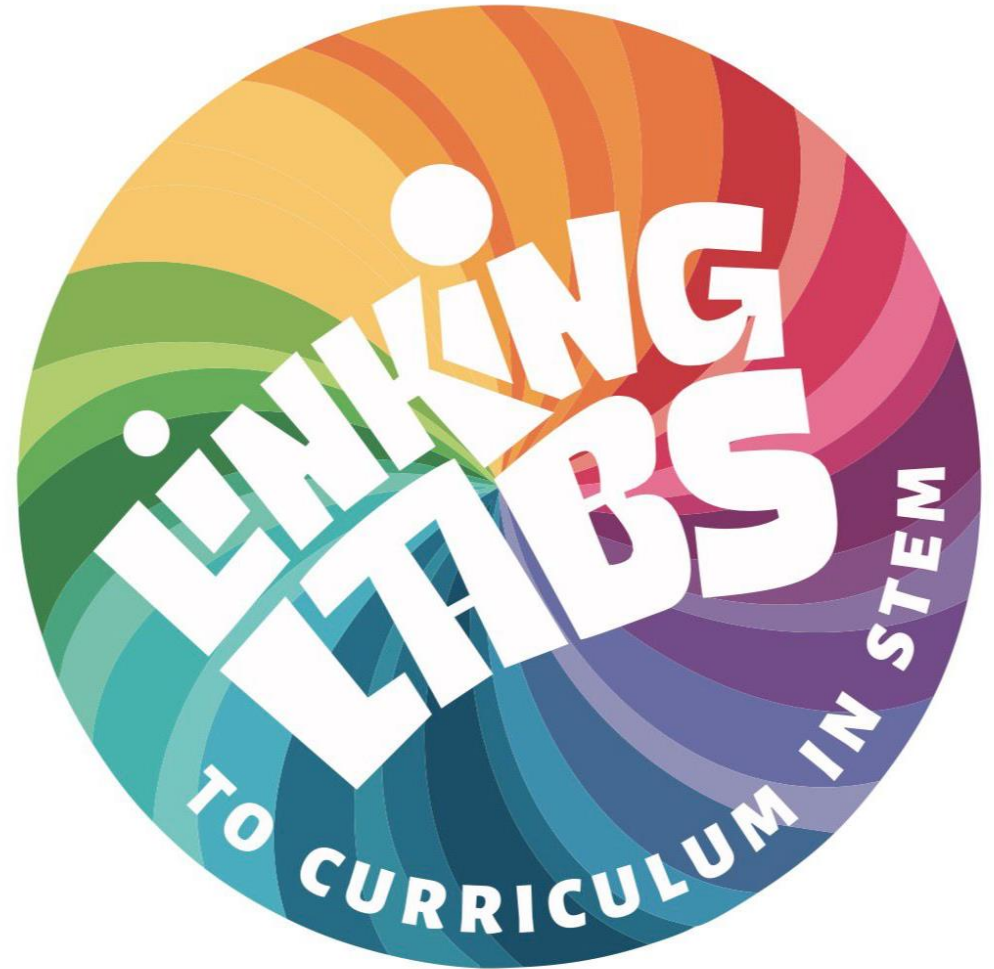
The Scientific Method

Do we remember the steps?
We start by.....



Today's agenda

- First activity: What are these folks doing?
- Second activity: "ouch I cut my finger", what is going to happen to me??
- Third activity: Building a rainbow





@clipartkey.com

*Ouch I cut my
finger, what is going
to happen?!!*

*Am I going to have this
gap forever ??? Or it is
going to close ???*

Don't worry Felix !!



@clipartkey.com

The healthy skin cells
surrounding the wound
detect the gap and go
into **cell division** to close it

Process of cell division responsible for healing is called: Mitosis



Cells go into mitosis when we need to grow or to heal

Mitosis

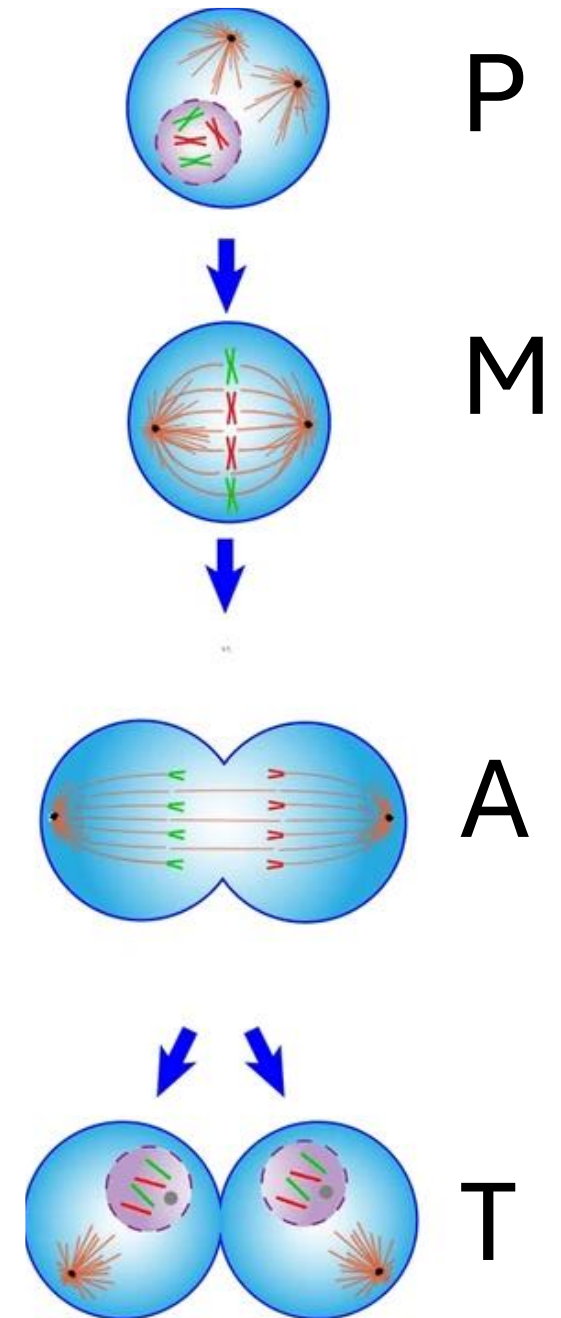
The Copy

Make an identical clone

Somatic cells

Process of Mitosis: Phases

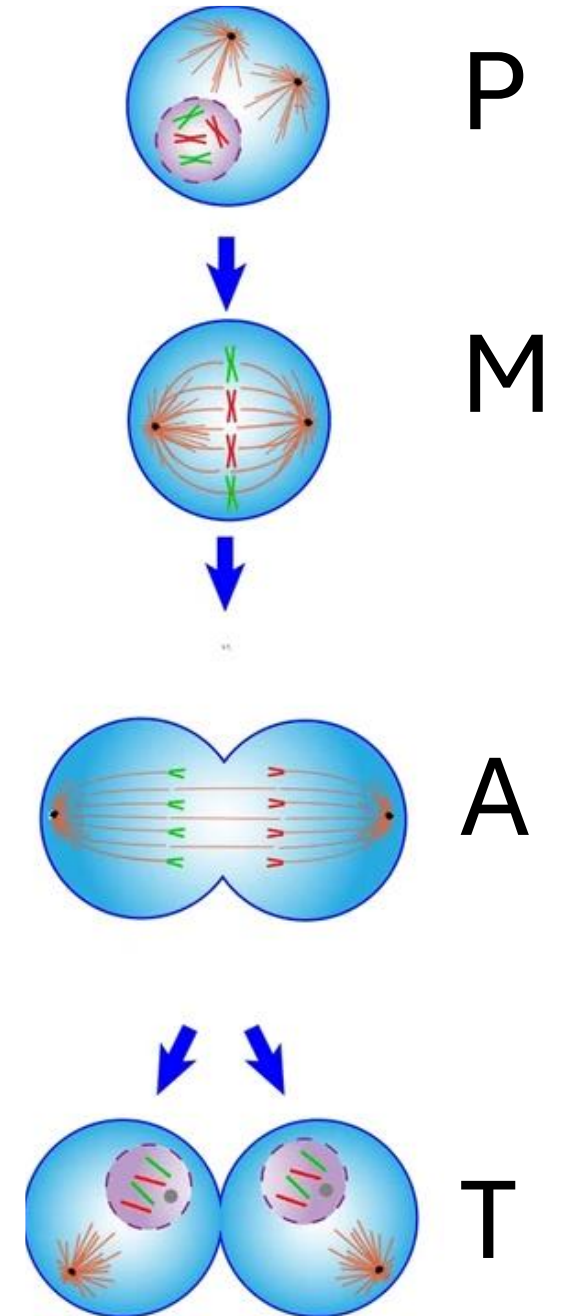
The "PMAT" Acronym (The Phases)



Process of Mitosis: Phases

The "PMAT" Acronym (The Phases)

Prophase (Prepare): The DNA "scrunches up" into visible chromosomes and the wall of the nucleus dissolves.

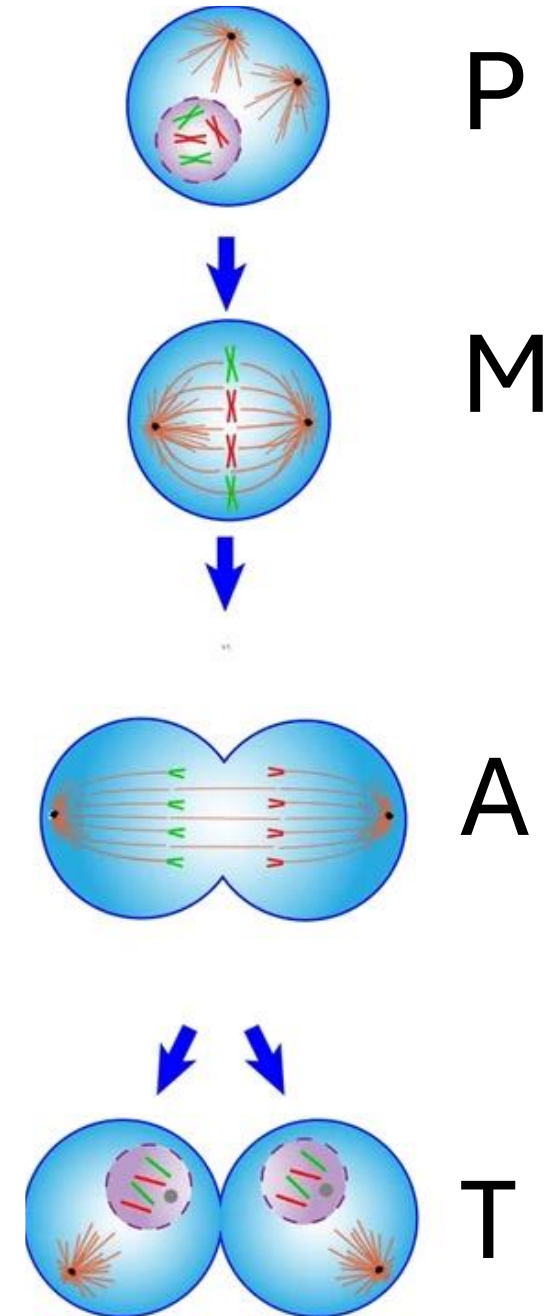


Process of Mitosis: Phases

The "PMAT" Acronym (The Phases)

Prophase (Prepare): The DNA "scrunches up" into visible chromosomes and the wall of the nucleus dissolves.

Metaphase (Middle): The chromosomes line up in a single file row right down the centre of the cell.



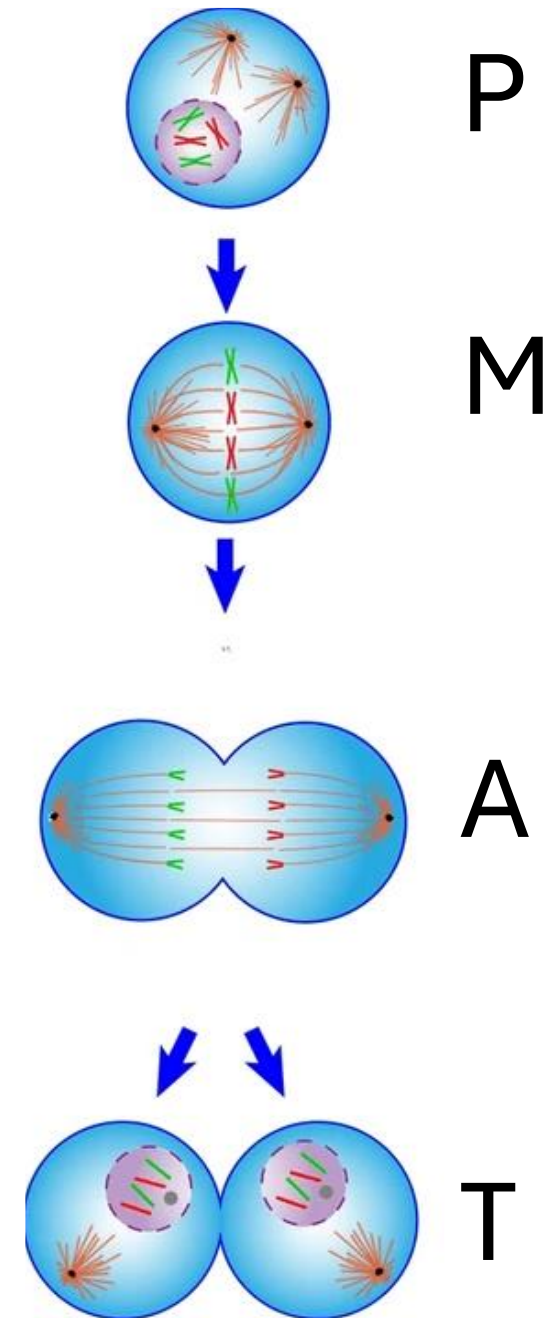
Process of Mitosis: Phases

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Anaphase (Apart): The sister chromatids are pulled away from each other toward opposite ends of the cell.



Process of Mitosis: Phases

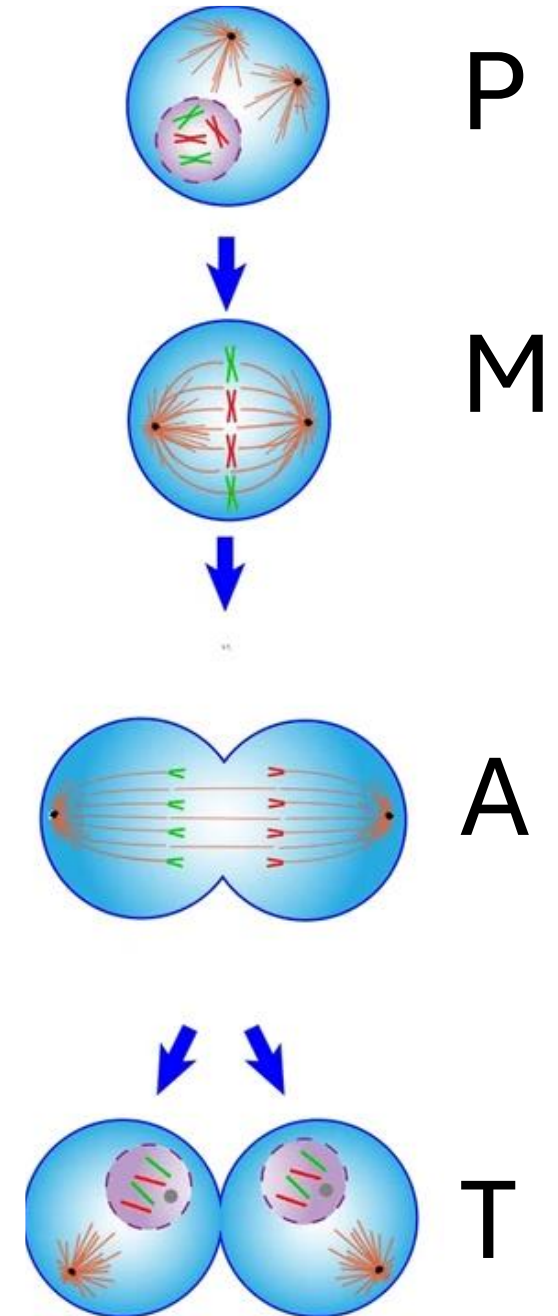
The "PMAT" Acronym (The Phases)

Prophase (Prepare): The DNA "scrunches up" into visible chromosomes and the wall of the nucleus dissolves.

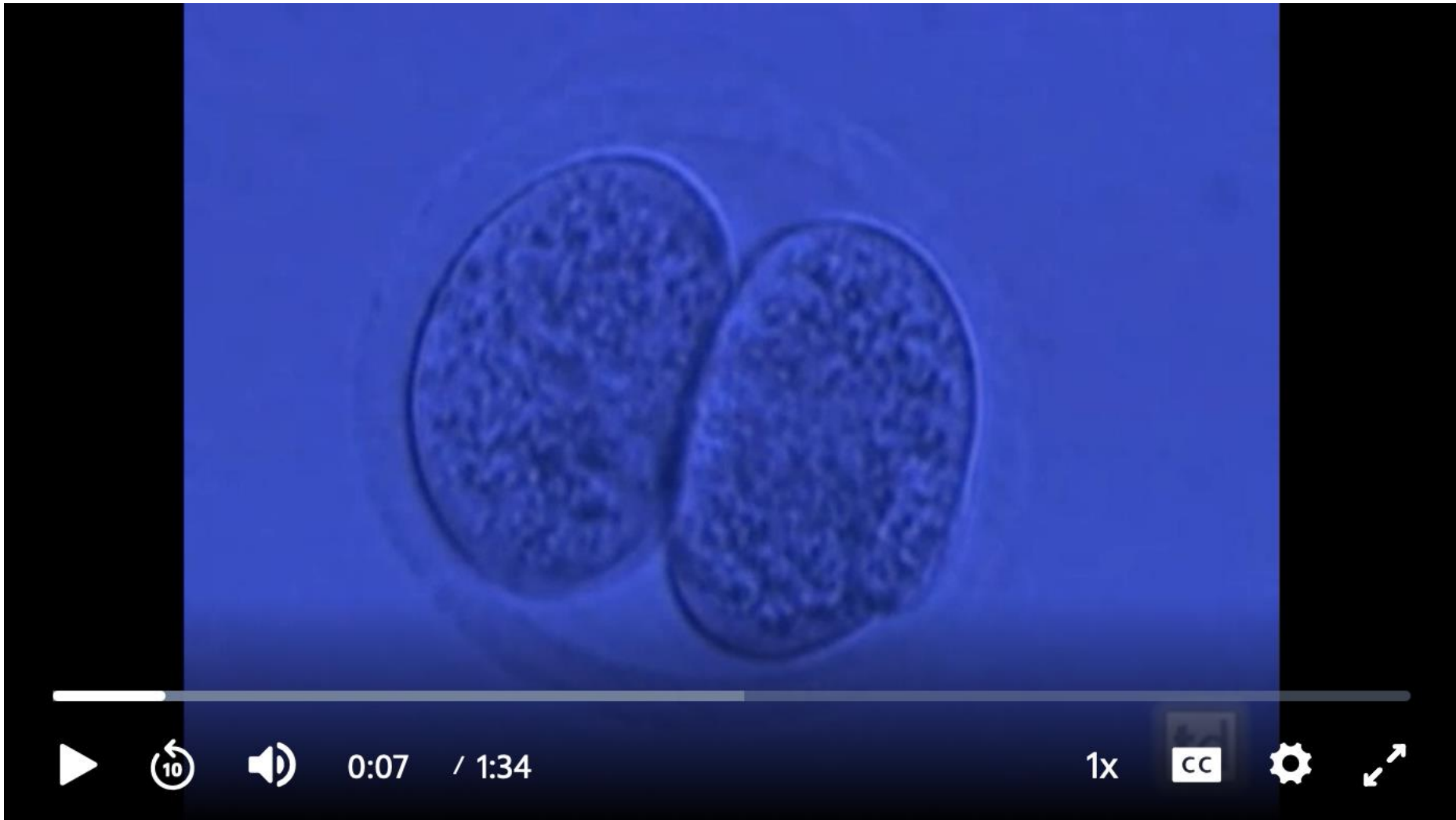
Metaphase (Middle): The chromosomes line up in a single file row right down the centre of the cell.

Anaphase (Apart): The sister chromatids are pulled away from each other toward opposite ends of the cell.

Telophase (Two): Two new nuclei form around the separated DNA. The cell starts to pinch in the middle.

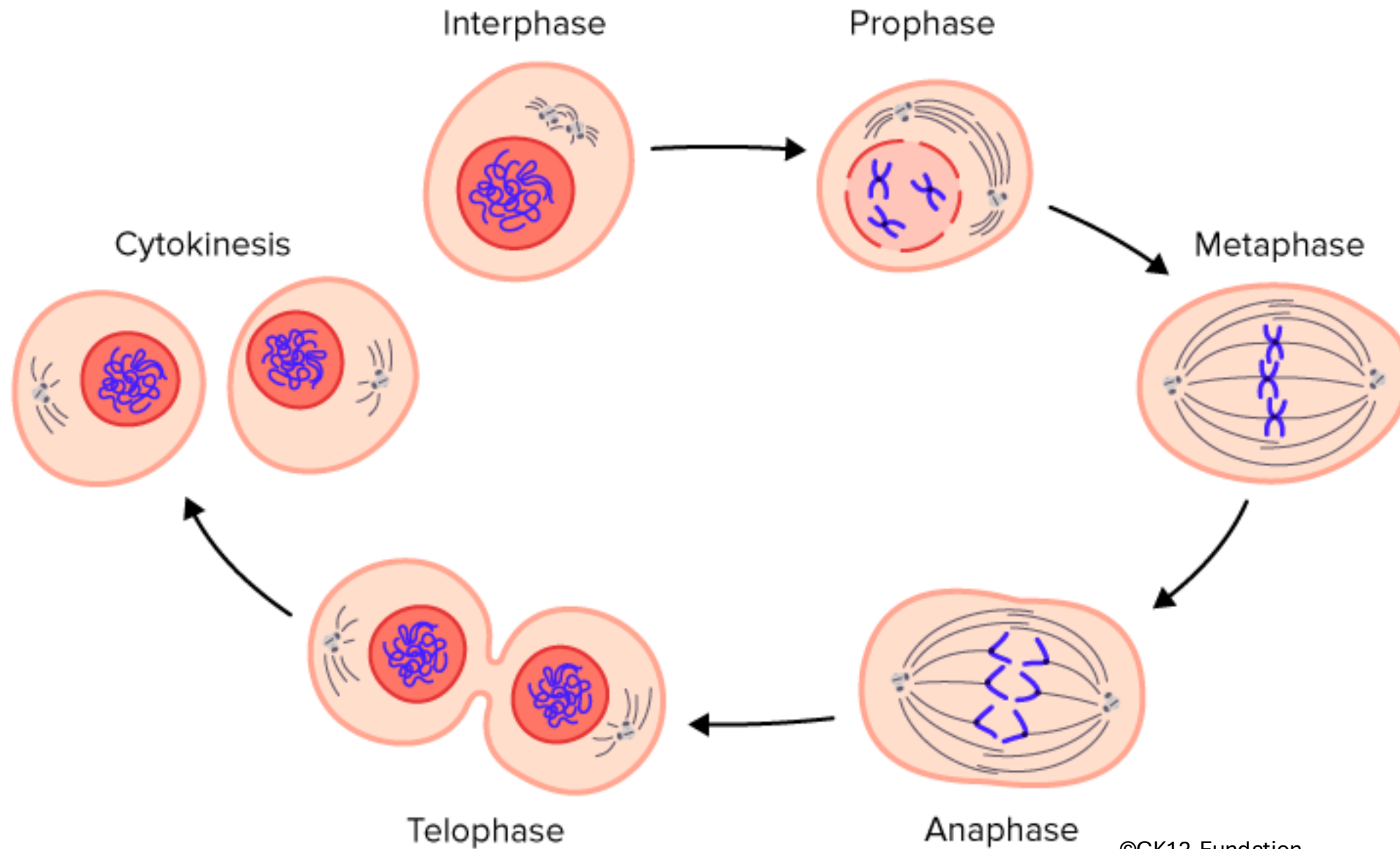


Process of Mitosis



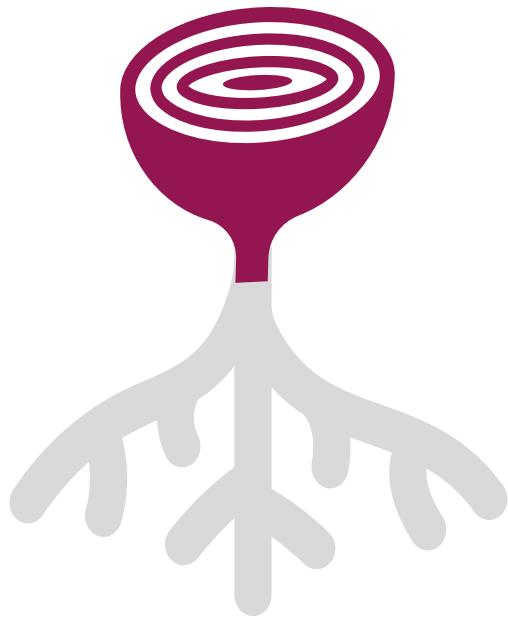
<https://www.pbslearningmedia.org/resource/tdc02.sci.life.stru.dnadivide/mitosis/>

Mitosis is only one part of the cell cycle

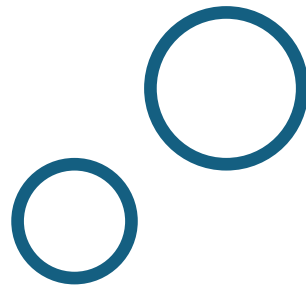


©CK12-Foundation

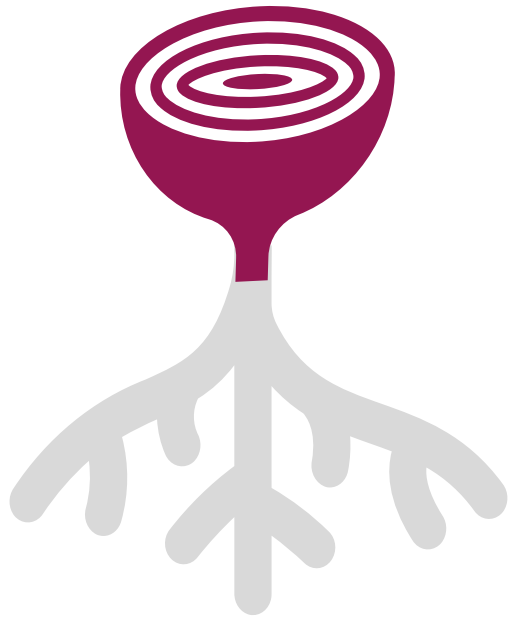
Today in the lab: close look at the process of Mitosis using onion roots



Any idea why the root
and not the layers of the
onion.....



Today in the lab: close look at the process of Mitosis using onion roots



We are going to try to
answer which stage the
cells spend more time in.

Laboratory safety

You will need to use:

Lab coat

Gloves

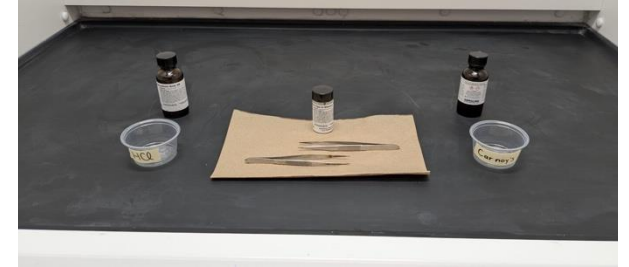
Safety goggles

Laboratory setting



Laboratory procedure

1) Using forceps, get an onion root from the "Carnoy" cup (in fume hood) and place it inside your plastic cup

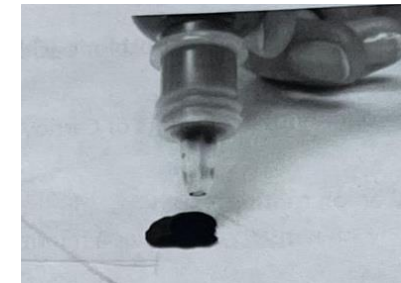


2) Once at your bench, place the onion root on a slide using forceps

3) Cut 1 to 2mm of the root tip on top of the slide

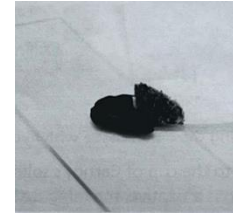


4) Add 1 drop of Toluidine blue (TBO) on top of the root and wait 2 minutes



Laboratory procedure

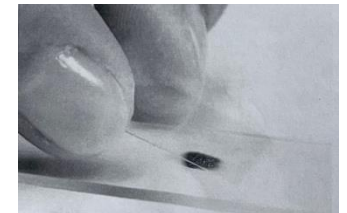
5) Wait 2 minutes and then blot away the stain with paper

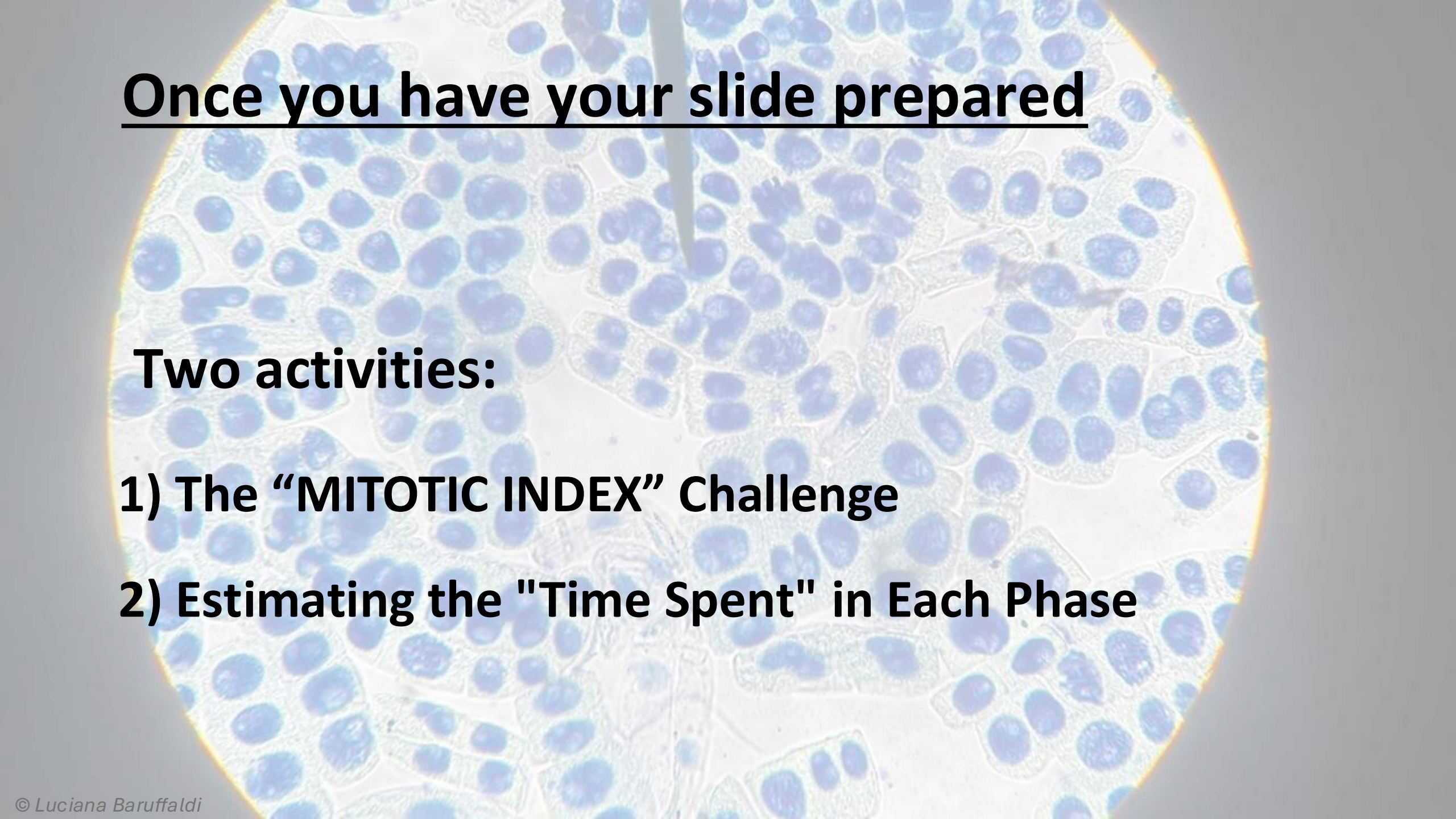


6) Cover the root tip with 1 drop of water



7) Place a coverslip over the root tip and apply even pressure to crush root tip into a flat, even level of cells



A circular inset showing a microscopic view of onion skin cells. The cells are roughly rectangular and arranged in a brick-like pattern. Each cell has a large, dark blue nucleus and a lighter blue cytoplasm. The cell walls are clearly visible as thin, light-colored lines.

Once you have your slide prepared

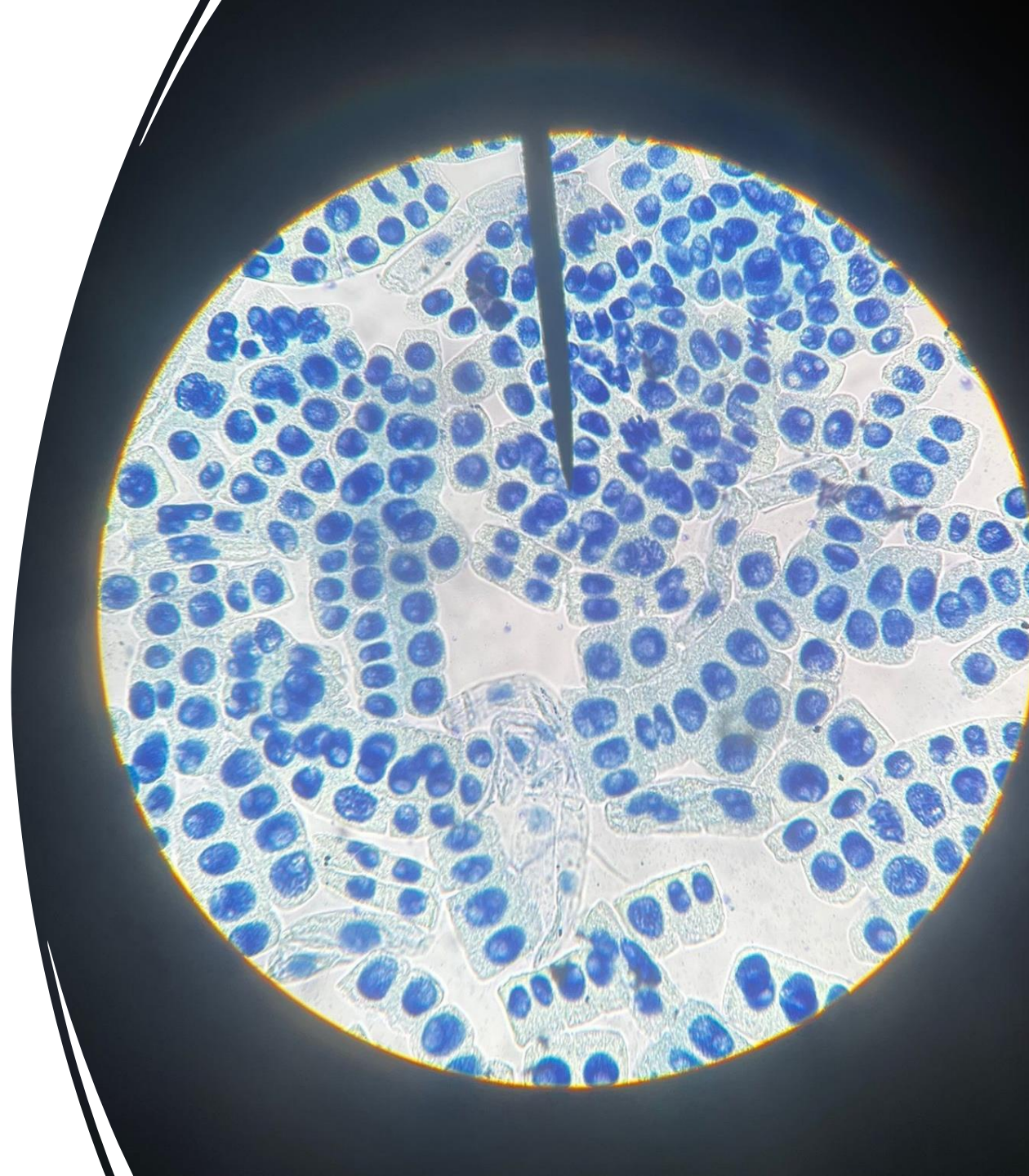
Two activities:

- 1) The “MITOTIC INDEX” Challenge**
- 2) Estimating the "Time Spent" in Each Phase**

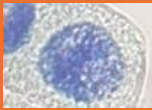
1) The "Mitotic Index" Challenge

The Task: Count 50 cells in YOUR field of view and categorize them by their stage of the cell cycle.

The Data: Record how many cells are in Interphase vs. Prophase, Metaphase, Anaphase, and Telophase.

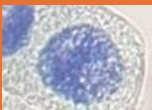


Make this table in your notebook

Stage		Tally (Total of 50 cells)	Percentage (%)
Interphase			
Prophase	 		
Metaphase	 		
Anaphase	 		
Telophase			
TOTAL		50	100%

$$\text{Mitotic Index} = \left(\frac{\text{Number of cells in mitosis}}{\text{Total number of cells observed}} \right) \times 100$$

Enter class data

Stage		Percentage (%)
Interphase		
Prophase	 	
Metaphase	 	
Anaphase	 	
Telophase		
TOTAL		100%

2) Estimating the "Time Spent" in Each Phase

Use your **Mitotic Index** data, and turn those percentages into a time-based model.

Assumption: An onion root cell takes **24 hours** to complete one full cycle.

Pie Chart or a **24-hour Clock** showing the "Life of an Onion Cell"

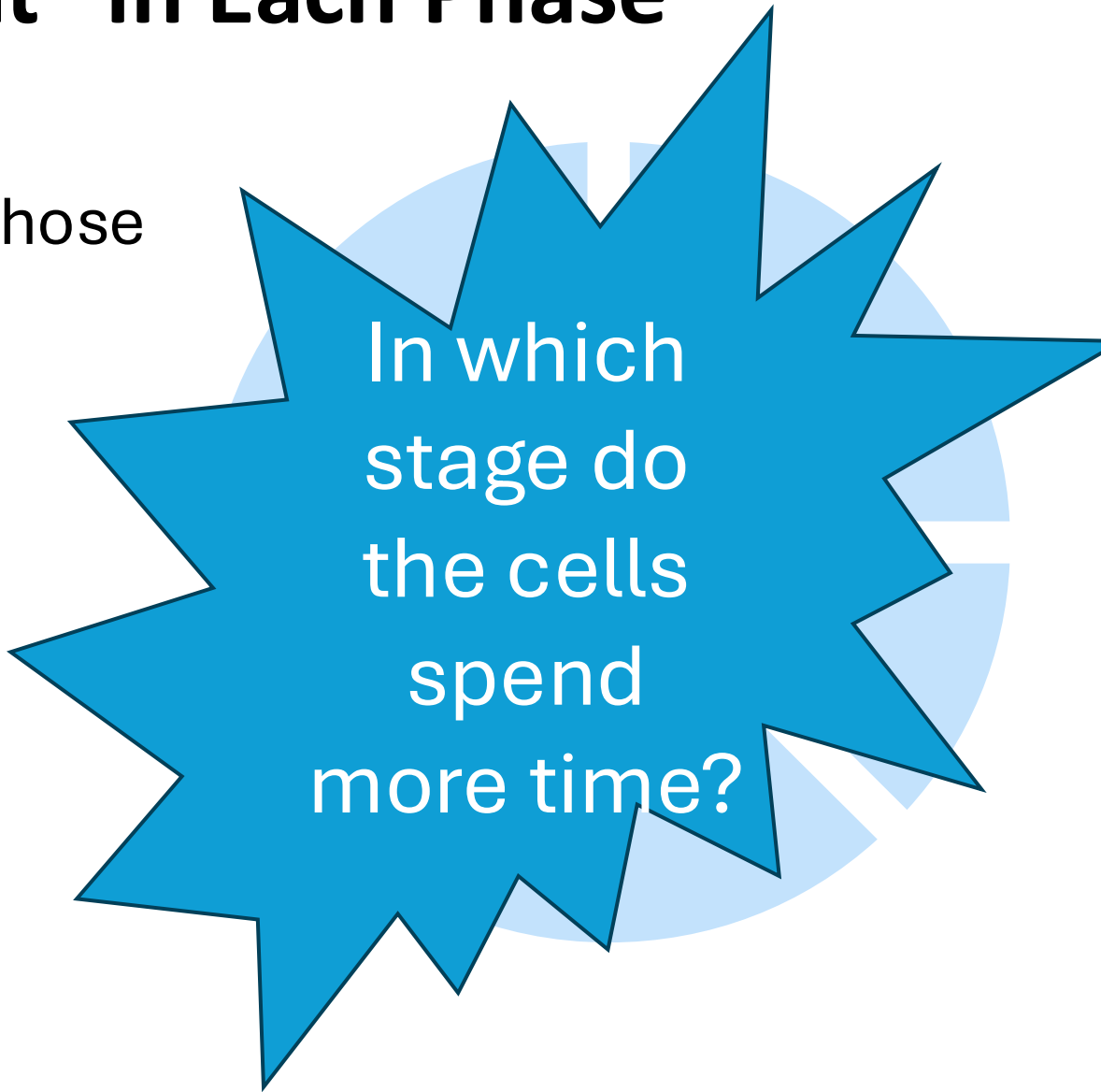


2) Estimating the "Time Spent" in Each Phase

Use your **Mitotic Index** data, and turn those percentages into a time-based model.

Assumption: An onion root cell takes **24 hours** to complete one full cycle.

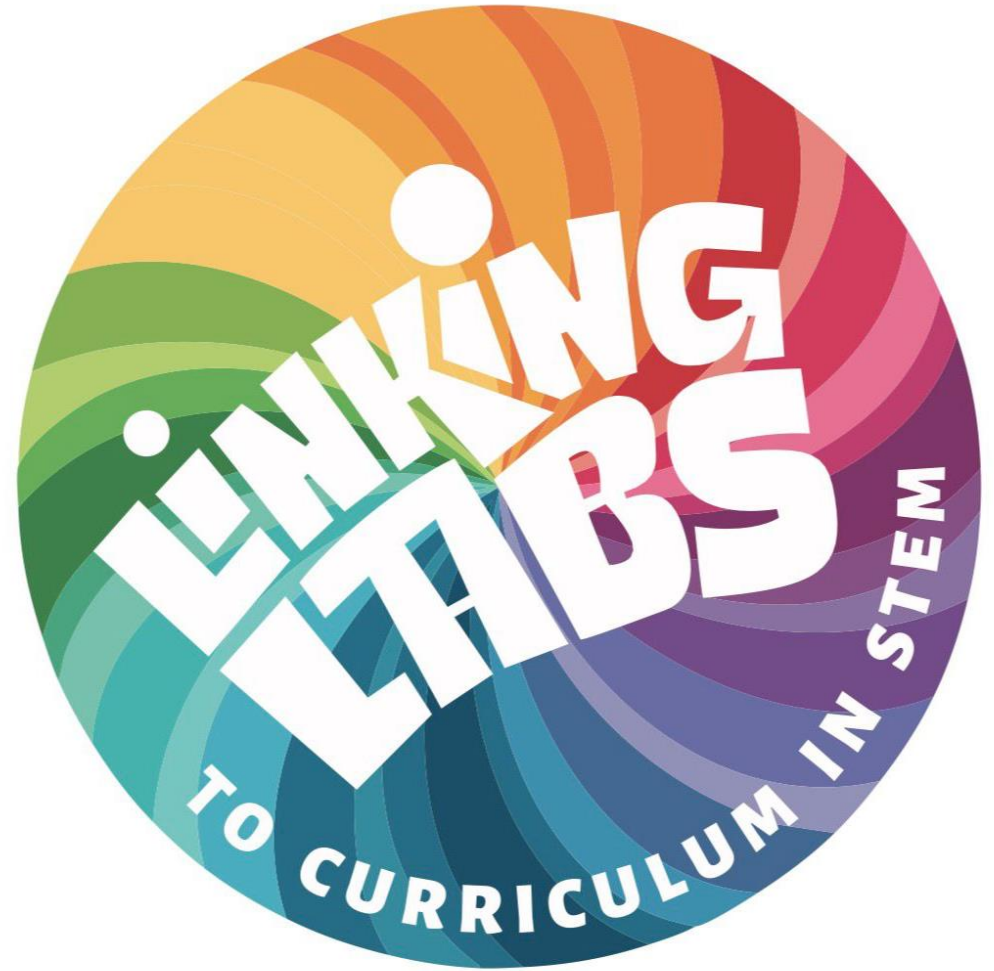
Pie Chart or a **24-hour Clock** showing the "Life of an Onion Cell"



In which stage do the cells spend more time?

Today's agenda

- First activity: What are these folks doing?
- Second activity: "ouch I cut my finger", what is going to happen to me??
- Third activity: Building a rainbow

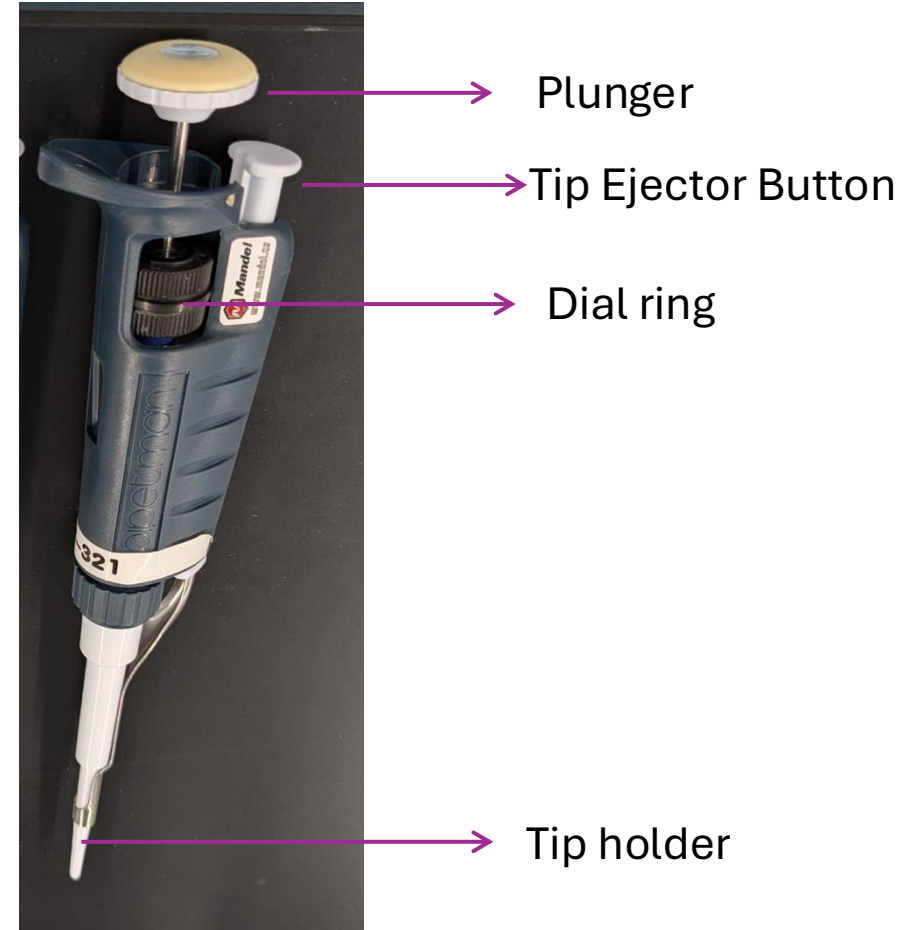


We are going to mix colors using micropipettes

Laboratory tools used to measure
specific volumes for experiments



very precise and can dispense
extremely small volumes



How to use micropipettes



→ Plunger

- The plunger is used to draw in and dispense solutions
- The plunger has 2 stopping points
- Try this out!

How to use micropipettes



→ Dial ring

The dial ring can be used to set your desired volume.

→ Tip holder

Attach the pipette tip for the micropipette (there are different sizes).

How to use micropipettes



→ Tip holder

→ Tip



Tip

*How to use **micropipettes***



- Push the plunger to the first stop
- Then place the micropipette tip into the solution
- Now slowly release the plunger to draw in solution
- Once there is solution in the tip DO NOT invert the micropipette, always keep it up right

How to use micropipettes



- To dispense the solution gently press the plunger to the first stop, then to the second stop
- The purpose of the second stop is to dispense any remaining solution in the tip.

How to use micropipettes



Tip ejector

- Hold the plunger at the second stop till the pipette tip is out of the microfuge tube.
- This prevents accidentally drawing the solution back into the tip.
- Then the last step is to eject the used tip into the waste bucket by pressing the tip ejector button

How to use micropipettes



- Choose the correct micropipette
- Set your desired volume
- Attach the pipette tip
- Press down to the first stop BEFORE entering the solution
- Release the plunger to draw in solution. DO NOT invert the pipette
- In a microfuge tube press down to the FIRST stop, then the SECOND stop.
- Keep the plunger pressed till you exit the microfuge tube.
- Finally, eject the use tip!

P1000

P200

P20

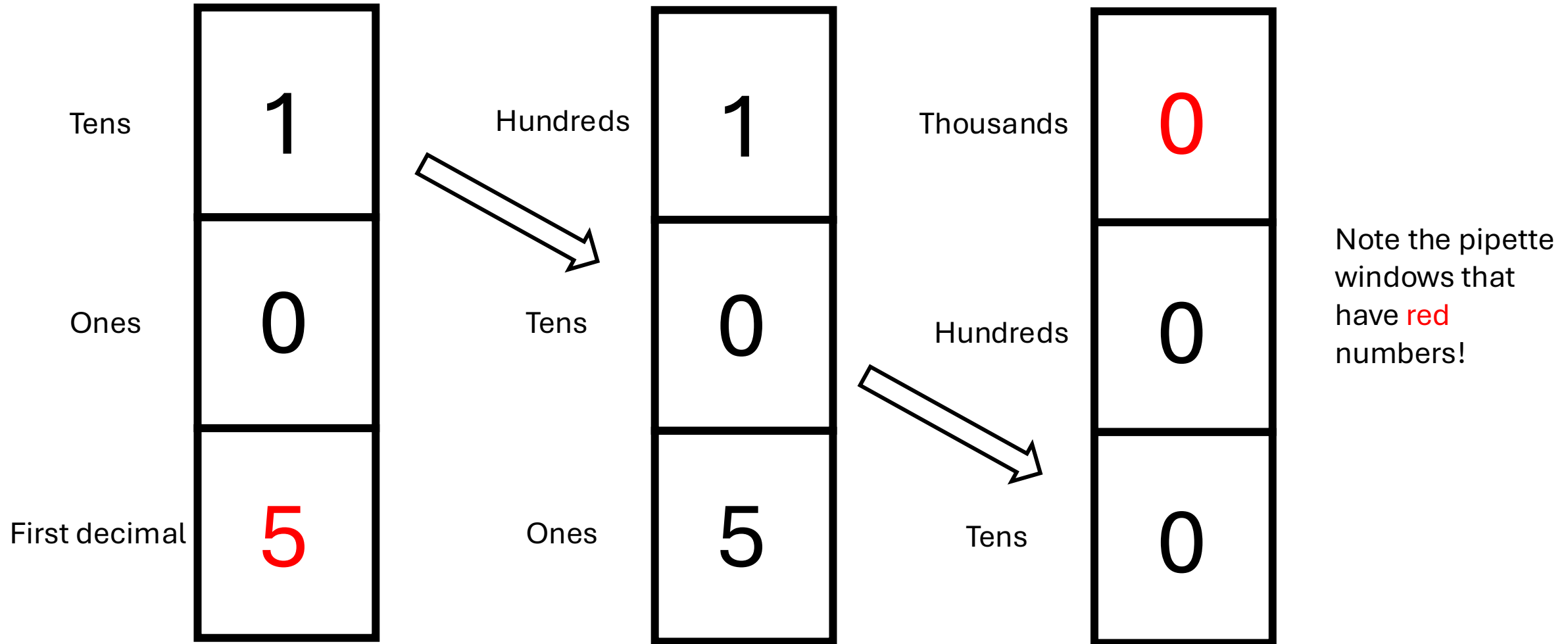


Reading the pipette window

P20 (2 – 20 μ L)

P200 (20 – 200 μ L)

P1000 (100 – 1000 μ L)



P1000: Used to measure
100 μ L to 1000 μ L

1000 μ L = 1mL

What volume is this pipette
set to?



In this picture this is 360 μ L

P200: Used to measure 20 μ L to 200 μ L

1000 μ L = 1mL

What volume is this pipette set to?



In this picture this is 44 μ L

P20: Used to measure 2 μ L
to 20 μ L

1000 μ L = 1mL

What volume is this pipette
set to?



In this picture this is 8 μ L

What micropipette will you use to dispense 700 μ L?



What micropipette will you use to dispense 700 μ L?



P1000 μ L

What micropipette will you use to dispense $8.5\mu\text{L}$?



What micropipette will you use to dispense 8.5 μ L?



P20 μ L

What micropipette will you use to dispense 72 μ L?



What micropipette will you use to dispense 72 μ L?



P200 μ L

What micropipette will you use to dispense 150 μ L

P200 μ L or P1000 μ L

Which do you think is more accurate ?



How to use micropipettes



- Choose the correct micropipette
- Set your desired volume
- Attach the pipette tip
- Press down to the first stop BEFORE entering the solution
- Release the plunger to draw in solution. DO NOT invert the pipette
- In a microfuge tube press down to the FIRST stop, then the SECOND stop.
- Keep the plunger pressed till you exit the microfuge tube.
- Finally, eject the use tip!

Micropipetting



Laboratory setting



Today's Task! Are you ready to Pipette?

Table 1: Volumes for addition to each tube

	Tube A	Tube B	Tube C	Tube D
Blue Stock	85	360	450	0
Green Stock	0	0	0	380
Red Stock	440	440	82	0
Yellow Stock	360	90	360	530

Table 2: Volumes for addition to each tube

	Tube A	Tube B	Tube C	Tube D
Blue Stock	8	36	45	0
Green Stock	0	0	0	38
Red Stock	44	44	8	0
Yellow Stock	37	9	36	52

When you are done coloring check
your colors with our team!!

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Production of the video: Luciana Baruffaldi and Alexandra Fernandez

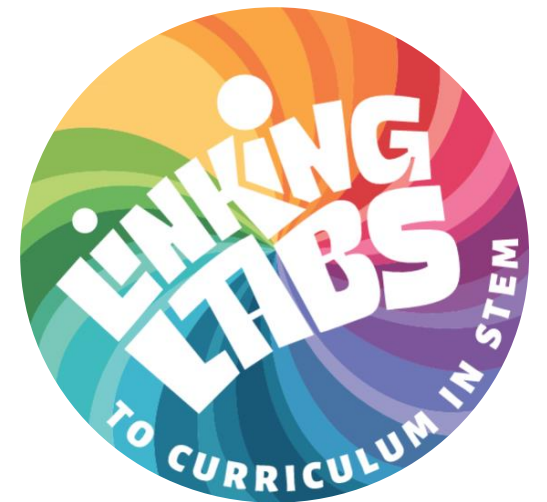
Directing and editing of the video: Alexandra Fernandez

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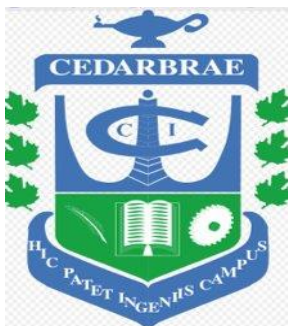
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Cedarbrae C.I



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